

Course Notes — SESSION 2

INFORMATION PROCESSING

1. Introduction

In our previous session, we characterized learning in AI systems and organizations as a process of using feedback to assign credit to problem-solving actions. We saw how assigning credit becomes challenging as a problem grows more complex. Finally, we viewed the complexity of the credit assignment problem under two AI paradigms — as the *combinatorial complexity* of actions to explore or exploit (reinforcement learning), or as the *complexity of managing* how actions are organized (learning in a society).

This discussion of credit assignment and complexity was evocative but imprecise. We talked in intuitive terms, such as a mouse going left or right in a maze, or managing various tasks for building this maze. To establish credit assignment as a basis for studying foundational AI ideas for the rest of the course, we need to be clearer about what we mean by a “**complex**” problem. In this session, we will define complexity as feedback required to learn to solve a problem that we can actually measure, or at least talk of, in formal (mathematical) terms. Our goal will be to think of **feedback as something like quantifiable units** akin to atoms or molecules.

By convention, we will from now on usually refer to feedback as “**information**”. This term “information” will flexibly stand in (for now) for any other term that we might think of for feedback, such as data, sense perceptions, or knowledge. Defining what we mean by information, however, will require precise assumptions. If information units are our building blocks, our assumptions about them will shape how we think about the very nature of credit assignment. It may be tempting just to try to define information using the most basic units we can imagine, similar to atoms (or subatomic particles) in physics. Yet consider how biologists study entire cells as the building blocks of living systems, and are less interested in sub-cellular units such as atoms. Likewise, the question that we really need to ask is: *what units of information will be most helpful for us to study processes of intelligently learning to assign credit in AI systems and organizations?*

To explore this question robustly, we will, as in our last session, consider two paradigmatically-different approaches to information in the study of AI systems and organizations. One approach, called **Shannon information**, after its pioneer Claude Shannon (1916-2001, a co-founder of the AI field from MIT), measures information as **meaningless units** called **bits**. The *reinforcement paradigm* that we looked at in the previous session, along with other types of machine learning that we will look at later, are consistent with Shannon’s approach. A different approach, which we will call **Simon information**, after Herbert Simon (1916-2001, also a co-founder of the AI field, from Carnegie-Mellon University) instead measures information as **meaningful units**

called **lists**. The *paradigm of learning in a society* from last session, along with many other AI ideas outside of machine learning, are closer to Simon's approach.

2. Establishing some shared terminology (and convenient fictions)

Before diving into the two approaches to information, let us establish some shared terminology. This shared terminology will empower us to think about credit assignment in precise terms across diverse AI ideas. As you will see, however, it will also require us to maintain convenient fictions about information that we will need to keep carefully in mind.

First, by **information**, we will refer not just to numerical data, but to any types of **symbols** for describing physical objects in, or concepts about, the world (i.e., problem-solving environments). Symbols may include **alphabetic characters, words, images, videos, sound waves** or any other observations that can be recorded. A pilot, for example, receives feedback as symbolic information in the visual displays of cockpit instruments, audio updates from air traffic control, or ear pressure from a sudden change in altitude. Symbols may be received either from another agent (human or artificial), or just by interacting with the physical world (e.g., gravity).

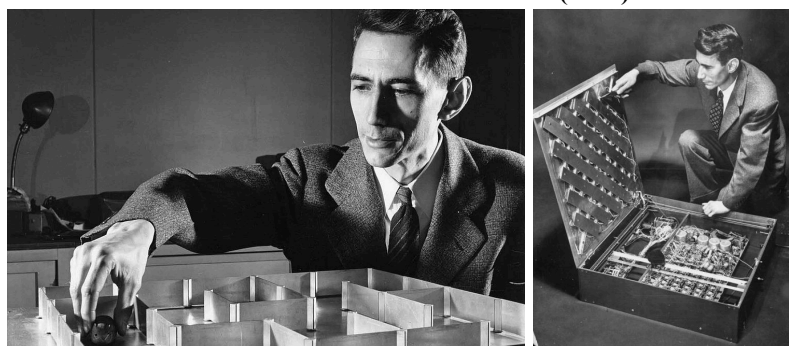
Second, we will refer to any use of information to learn to assign credit to a problem as “**information processing**” (indeed, an early proposal was to name the field “complex information processing” rather than AI). By information processing, we will refer broadly to any actions for “storing, manipulating, displaying, and otherwise trafficking in information” (Cantwell-Smith 2021) by **agents** (any entities taking goal-directed actions) interacting with **problem-solving environments** (anything external to agents that is also relevant to a problem). For the pilot, such actions could be physical movements (e.g., pulling on the throttle in the cockpit), choices or decisions (e.g., switching from autopilot to manual mode), or simply observations (e.g., checking the altitude, air pressure, or remaining fuel).

Third, we will view information processing as **acts of communication among agents**, and AI systems and organizations as communication systems. What agents do to be intelligent is essentially “package” information into “**messages**” that they **send**, or **receive** and evaluate messages from other agents. These actions are collectively referred to as “**message passing**”, such that AI systems and organizations can also be called “message passing systems”. This view of information processing as communication requires maintaining our first convenient fiction, which is that an *agent's problem-solving environment is simply other agents who process information*, even if some or most of the environment is really not agentic (e.g., the atmosphere that the pilot interacts with). This fiction will allow us to talk about agents and environments as a single system in which messages are passed back and forth and, hence, allow us to formalize ways of talking about the complexity of the system.

Shannon, whose approach to information we will describe momentarily, famously displayed this convenient fiction in a robotic mouse he built in 1950 called **Theseus**, which “learns” to navigate a maze whose walls can be flexibly configured (as depicted on the left). Theseus’s maze “environment” is really a network of electrical circuit “agents” when you open it up (as depicted on the right). When Theseus takes actions in a part of the maze, the relevant circuits “send messages” to him about whether a particular path in the maze is open or blocked.

Conjuring an agent’s problem-solving environment in this way — as communicating agents akin to a **circuit network** — has been a dominant perspective in the AI field since it emerged in the 1940s and 1950s (when designing circuits for telephone networks was a canonical example of a complex problem), and on up to today (where the canonical example is designing the networks of circuits in semiconductor chips). It equally has led to theories of organizations as essentially communication systems designed to structure flows of information. Though this characterization helps us define information about a problem, keep in mind that it also **blurs the distinction between an agent and its environment**. For example, in Theseus, the mouse’s “information processor” is really in the maze circuitry. This agent-environment distinction will not matter so much for the purposes of this session, but we will need to think carefully about it later on.

Claude Shannon’s Theseus robot (1950)



Fourth, we will (for now) refer to all basic actions underlying information processing or message passing as “**symbol manipulation**” operations. Such operations refer to diverse actions, ranging from applying arithmetic or algebraic rules, to using formal logic, to moving icons on a laptop screen, to speaking in natural language, to writing a prompt in a GenAI tool. Common to all such symbol manipulation operations, however, is that **learning to assign credit** arises from *evaluating information in patterns of the symbols*. Such evaluation may involve either **pattern matching** (matching exact symbols) or **pattern recognition** (matching similar symbols)).

Note that this symbol manipulation assumption will require us to maintain a second fiction, called the “**digital abstraction**”, *which is that we can talk about information as a symbolic thing, independent of the physical world*. Consider that Shannon’s Theseus depicted above has no overt symbols — the maze is just some electromechanical circuits connected by physical wiring, such

that “messages” passed from the maze to Theseus are really analog electrical pulses. The digital abstraction, however, is convenient (if we are careful about using it) by enabling us to relate any actions (analog or otherwise) to a single, coherent system for evaluating and transforming **patterns of symbols** into other patterns and, hence, for talking about information as a formalizable quantity that varies based on the complexity of a problem.

Fifth and finally, leveraging this convenient fiction, we will think of the **complexity** of a problem as the amount of symbol manipulation actions for solving it — or, more precisely, the complexity of the symbolic patterns that need to be matched or recognized to solve a problem. In AI, this quantity is often analogized to how much “**energy**” or “**work**” is required for solving a problem intelligently. Hence, we will say that processing more complex information requires greater “**processing power**” or “**processing capacity**”. For example, solving the problem of downloading the information in a video file of a lecture typically takes more “capacity” from your computer than an audio file of the same lecture, since the video file has more information (e.g., the file size is much larger).

With this terminology (and convenient fictions) established, we can now frame the general intelligence problem of credit assignment (for now) a bit more formally — in information processing terms or, equivalently, as a problem of message passing (of communicating patterns in symbols) among agents:

Assigning credit is an information process of evaluating (i.e., matching or recognizing) patterns in symbols about problems. Challenges of assigning credit arise from the need to evaluate more complex symbolic patterns.

Further, now we can frame a bit more formally what is meant by paradigmatically-different approaches to assigning credit in intelligent systems:

AI ideas differ in the symbolic patterns that they take to be the basic units of information and, as a result, in how they conceive of the capacity (i.e., complexity) required to process these patterns.

3. Approach 1: Shannon information (syntactic information processing)

In a 1948 paper, Shannon introduced an approach to thinking about information that has been so influential to the study of AI and organizations (and many other scientific and engineering fields) that it is still often simply called “**information theory**”. He developed his approach in the 1930s and 1940s, ostensibly to solve problems of routing calls over complex telephone networks while working at Bell Labs, the research lab of American Telephone & Telegraph Company (AT&T), which was the largest telephone operator in the US at the time. Yet Shannon had in mind a

general theory of communication in any system which, as he demonstrated with his 1950 robot Theseus, could in fact be used to develop machines that could in some sense be said to “learn”.

We will understand the Shannon information approach in terms of three assumptions: information processing as **syntactic**; units of information as **meaningless bits**; and processing information to learn to assign credit intelligently as ultimately a matter of **brute force**.

3.1 Learning to assign credit based on syntactic information processing

Shannon’s approach is based on his insight that, in an electrical engineering sense at least, the **semantic content** (the meaning) of messages sent over a telephone network (or any other communications system) is **irrelevant**. AT&T’s real problem was simply to ensure that information about the **syntactic structure** of symbolic patterns in messages is sent accurately. For example, when a pilot sends a message to air traffic control about the plane’s altitude, the altimeter reading needs to be accurate in the sense of the sequence of numerical symbols (e.g., if the altitude is 30,000 feet, ensuring that the reading is a “3” followed by four zeros). This syntactic information about the altitude is separate from whether the pilot or air traffic control interprets the meaning of 30,000 feet as too low or too high.

In the context of AI, Shannon’s key insight was that, remarkably, intelligently learning to assign credit might be thought of as similar to how AT&T needed to route calls in a telephone network — an intelligent agent needed only to process syntactic information to learn to solve a complex problem. His Theseus robot, for example, could “learn” to navigate any configuration of the maze by using “messages” sent by the maze circuits to assign credit to certain paths as good or bad, and then to modify its selection of sequences of actions over time. Such “learning” did not require Theseus to interpret the sequence of actions he took, other than associating the actions with good or bad paths, similar to the reinforcement paradigm that we studied last session.

3.2 Units of information: bits

Shannon proposed binary digits — 0’s or 1’s — that he called “**bits**” as the building block units of syntactic information processing. Bits are linked together into “messages” as **patterns** of 0’s and 1’s called **bit strings**. All symbols can be reduced to bit strings. Converting symbols into bit strings is called **encoding** a message; reconstructing the symbols in the original message is called **decoding** it. For example, a standard way of encoding the letter “A” in a computer is as the bit string “01000001”. Since messages often have redundant sequences of symbols (e.g., repeating the same words or grammatical patterns), the encoded bit strings in them are not a true measure of the number of bits required to communicate the message. What counts is how efficiently the message can be communicated after statistical redundancies are removed, a process that allows the original message to be “**compressed**” into a smaller number of bit strings that, in turn, reduces the processing capacity required.

The amount of *information* in a message, however, is *not* measured simply by counting the number of compressed bits. For Shannon, this amount instead refers to the **uncertainty reduced** about the probability distribution of outcomes from receiving a compressed message. This uncertainty reduction depends on (1) the complexity of the problem, or how many possible “messages” *could* be received (e.g., how many sequences of interdependent actions Theseus *could* take in a maze), and (2) the perceived **probability** of a message by the agent, given what the agent has already learned (e.g., if Theseus “exploits” similar sequences of actions to what he already tried, the messages that he “receives” from the maze will give him less information than if he explores new sequences).

A crowning achievement of Shannon is to formalize a measure of uncertainty about the statistical distribution of the probabilities of messages as a quantity that he called “**entropy**”, named for its similarity to entropy in thermodynamics. Shannon entropy can be thought of as the information in each possible bit string message (e.g., each possible sequence in Theseus’s maze) weighted by the probability that the message occurs (e.g., how likely Theseus believes that each sequence of actions is to lead to a reward). Hence, *the amount of information in a message is inversely related to its probability, with rare messages having more information than common ones* (hence, exploring new actions generates more information than exploiting existing ones).

In the language of our course, you can think of Shannon entropy roughly as a more precise way of measuring the **combinatorial complexity** of a problem by doing so **in probabilistic terms** (i.e., in terms of a statistical distribution of the probabilities of actions (messages)). Whereas combinatorial complexity captures simply the number of possible action sequences in a problem, entropy also accounts for how likely some action sequences are. In this sense, Shannon entropy measures not just the objective complexity of a problem, but *how an agent’s uncertainty about this complexity may be reduced as it “learns” by processing syntactic information about it*.

3.3 Credit assignment as a challenge of limited “channel capacity”

What do Shannon’s assumptions about the units (bits) and complexity (entropy) of information imply about the general problem of intelligence in AI systems and organizations — the problem of learning to assign credit? Overall, the implication is, for solving any particular problem, an agent’s ability to learn to assign credit is *entirely a matter of their capacity for processing syntactic information (i.e., of the capacity to evaluate patterns in bit strings that encode symbolic patterns in messages)*. Hence, you can think of all AI ideas consistent with Shannon information as various ways of dealing with **limits to syntactic processing capacity**.

Shannon described these limits as arising from how communication, while based on syntactic symbols, still ultimately happens over **physical channels**, or restricted pathways to send and receive messages, such as AT&T’s telephone network or fiber optic cables today. Beyond electronic or digital equipment, channels can refer to any way of restricting the processing of

information. The structure of an organization, for example, is a channel in the sense that it controls who can communicate with whom in the organization (Cyert & March 1955). A rigid hierarchy is typically thought of as channeling information in a more restrictive fashion (up and down) than a non-hierarchical organization (though with possible advantages of greater efficiency and control).

Limits to channel capacity mean that only a certain amount of information, or “**bandwidth**”, can be processed at a certain speed over any one period of time. A conventional TV channel, for example, has access to only a limited amount of bandwidth, while fiber optic cables in a 5G network have more capacity. Channel capacity limits mean that information in messages is what Shannon called “noisy” — that *the symbolic patterns encoded into bit strings are only communicated with a certain probability*. “**Noise**” (communication of wrong or irrelevant bit strings) is contrasted with “**signals**”, or the accurate communication of bit strings in a message. The **signal-to-noise ratio** is a measure of how reliable a message can be communicated.

A fundamental mathematical finding of Shannon’s is that, **as signal-to-noise increases, the amount of information that can be reliably transmitted increases combinatorially**. Hence, in theory at least, *a linear increase in channel capacity* may be enough to address the issue of combinatorial explosion (or, in information processing terms, of expanding “entropy”) from our last session. Hence, *Shannon information implies solving the problem of credit assignment in AI systems and organizations is ultimately a matter of increasing channel capacity*.

3.4 “Brute force” solutions to credit assignment

In 2019, **Richard Sutton**, whose ideas about reinforcement learning we looked at last session, wrote a short blog post titled “**The Bitter Lesson**”. His post went viral, and has come to epitomize a dominant view of AI today beholden to the Shannon information approach. For Sutton, the “bitter lesson” is the result of over seven decades of research and practice in the AI field (roughly since Shannon demoed his Theseus robot in 1950). This result, in his view, is that “progress” in the AI field has mostly been made not by AI researchers thinking of complex strategies for assigning credit, but rather of embarrassingly simple strategies (e.g., Sutton’s idea of a value function) paired with massive increases in channel capacity (exponential gains in semiconductor speed and data storage and availability).

In AI, having a system learn to assign credit intelligently just by pairing a simple strategy with essentially big computers is called a **brute force** strategy. You should see that a belief in brute force rests on a specific assumption in Shannon’s approach: that *any combinatorial explosion in the complexity of problems will be more than made up for by the exponential increase in signal-to-noise from just a linear increase in channel capacity*.

If the brute force assumption about AI is correct, then you should see that, given expected continued vast expansion in digital infrastructure, there are *few obvious limits to how the AI field*

will progress. Consider the estimate from some neuroscientists that the human brain has 100 trillion synapses. If all the brain is analogous to a Shannon information processor — if the functioning of the brain can be fundamentally understood as the processing of bit strings to send and receive messages among neuron “agents” — then it is straightforward to calculate how much information processing capacity would be required to construct an artificial brain. Given that such capacity is simply a matter of adding more digital infrastructure (i.e., more chips and data storage), then constructing an artificial brain is just a matter of time. This way of thinking has led some in the AI field to conclude that we will soon get a generally intelligent or superintelligent AI system, including fears that such a system poses an existential threat to humankind.

Before we move on, it is worth noting that Shannon himself would likely not have believed any of the above claims about brute force and intelligence. The ambitious term for his approach (“information theory”) was insisted on by his co-author; the term “entropy” was chosen also for rhetorical reasons, to make the theory itself sound complex and mysterious. Shannon viewed his approach to syntactic information processing as a powerful, but targeted solution for certain engineering problems, and as a way to start to think about how to design learning systems in the nascent AI field. Infatuation with Shannon information as a “theory of everything” in AI and science more broadly arose as early as the 1950s, shortly after his paper was published, in ways that mirror some of the rhetoric in the AI field today. Shannon himself wrote a commentary calling those who exaggerated the implications of his approach as on the “**information bandwagon**”, and cautioned for the need for diverse ideas for understanding intelligence.

4. Approach 2: Simon information (semantic information processing)

In a famous 1962 paper titled “The Architecture of Complexity”, Herbert Simon described an alternative to Shannon’s approach to understanding information and its relationship to intelligent learning. Simon developed his approach out of his research on decision-making processes in organizations from the 1930s to 1950s, including a field study of the budgeting bureau in a city government, and by observing corporate managers make decisions under experimental settings. Though ostensibly the research was to improve administrative decision-making, Simon had in mind a general theory of problem-solving, which he demonstrated in the mid-1950 using simulations of solving logic puzzles and playing chess (developed with his colleague, and fellow AI co-founder Allen Newell).

We will contrast Simon information from Shannon’s approach in terms of three assumptions: information processing as **semantic** and not just syntactic; the fundamental units of information for intelligence as **meaningful lists**, rather than meaningless bits; and intelligent information processing as ultimately a matter of **abstracting** a problem to **modularize** (i.e., decompose) it, rather than just a matter of adding channel capacity (i.e., brute force).

4.1 Learning based on semantic information processing

Simon's approach emerged from his insight that in organizational decision-making — or, it seemed to him, any complex system — **syntactic information processing** on its own would be *far too slow and inefficient* for intelligence to emerge, even if possible in theory. The real problem was instead to identify **semantic structure** in symbolic patterns that would allow information about complex problems to be decomposed into these structures. Consider our example from the previous session of a budgeting committee of a pharmaceutical company that needs to decide in which business units to invest. In theory, the managers on the committee could make their budgeting decisions by processing bit strings extracted from dense reports consisting of detailed numerical data on past performance and future sales forecasts. In practice, they rely on semantic categories, such as the names of the business units and the types of innovations that they propose, and summaries of numbers such as last year's sales figures.

In the context of AI, Simon's key insight was that such reliance on semantic categories was not just due to human managers' limited "channel capacity". Rather, how managers make complex decisions in such settings may be thought of as similar to how semantic information is fundamental to learning to solve any combinatorially complex problem in any intelligent system. Shannon's Theseus, under this view, "learned" without such categories since the maze was still relatively simple. Sutton's "bitter lesson", Simon might have said, can be explained by how AI technologies based on syntactic information processing (e.g., reinforcement learning) only "learned" when provided with a vast infrastructure of meaningful symbols (e.g., carefully designed programming languages, databases, software architectures, etc.).

In his 1962 paper, Simon famously illustrated the need for semantic information through his **watchmaker parable**. In Simon's parable, the problem is to learn to assemble a mechanical watch of 1,000 parts. There is a chance of the assembly task being interrupted, in which case any incomplete assemblies fall apart and the task starts over. Simon describes two watchmakers. One decomposes the watch into 100 named (i.e., meaningful) **modules** (i.e., components) of 10 parts each. The other assembles the 1,000 parts one at a time. Even assuming a modest chance of interruption of 1%, Simon calculates, the first watchmaker needs to repeat only a handful of subtasks per interruption, while the second requires repeating 100 subtasks on average, and around 4,000 times longer to complete the overall assembly task.

4.2 Units of information: lists

Simon proposed that, rather than reduce symbols to individual bits that are formed into meaningless strings of 0's and 1's, the bits should be organized into meaningful patterns (e.g., names of the components of watches) that he called **lists**. Lists differ from bits in that they are not just arbitrary strings of 0's and 1's — they have a **name**, followed by the "items" of the list. The name of the list is a label that "**designates**" the meaning of the symbols ("items") on the list — that is, the name indicates what the symbols externally refer to in the physical or conceptual

world that are relevant to a problem. Whereas bits are packaged into bit strings, lists can be “linked” together into *arbitrarily abstract patterns*, such as concepts, sentences, paragraphs, blocks of code, or equations. For example, while Shannon information encodes the letter A as the bit string “01000001”, Simon information might encode it as the list: (Letter A), where “letter” is the name designating a letter of the alphabet, and “A” is the “item” of the alphabet that is relevant here.

In their Turing Award speech (the equivalent to the Nobel Prize in computer science lecture), Simon and his colleague Newell explained this idea for list processing through what they called the “physical symbol system hypothesis” (1976). This is the hypothesis that “physical symbols” — symbols that meaningfully refer to the outside world, rather than just being stored as meaningless bit strings — are a “necessary and sufficient basis for intelligence” in an AI system or organization. The implication is that efficient intelligent systems arise not by sheer scaling of syntactic processing capacity, but by how meaningful abstractions of bit strings eliminate the need to process information about every detail of the environment. The key argument underlying the physical symbol system hypothesis is that, with a good enough set of ideas on how to develop abstractions (e.g., a good enough set of languages), agents can reason about any information in a meaningful way that, in turn, leads to efficient credit assignment.

To continue the comparison, whereas in Shannon information, the number of bits for learning to solve a problem refers to the required bits after *statistical redundancies* have been removed, in Simon information, such redundancies are among meaningful lists at various levels of abstraction and complexity. What matters, then, is not the total number of non-redundant lists per se (as in Shannon entropy), but rather the agent’s ability to describe the problem at a **higher-level of abstraction** and, hence, with fewer lists (i.e., describing a mechanical watch in terms of 10 components, rather than 1000 parts). A bit more formally, the key efficiency mechanism shifts from the **compression of bit strings**, to the **abstraction of list structures**.

4.3 Credit assignment as a challenge of limits to one’s abstractions

What do Simon’s different assumptions about the fundamental units of information (lists) imply about the general problem of intelligence in AI systems and organizations — i.e., the problem of learning to assign credit? In Simon information, the point is not really to have vast channel capacity for processing information. There is little interest in “noise” of information, or the “signal-to-noise” ratio. Processing information structured into lists that each **designate meaning in the world** implies that any noise will eventually be identified and eliminated. Agents do not have to just guess using probabilities to match patterns; they dynamically reason using their understanding of how information about the problem maps to the world.

Hence, the fundamental assumption in Shannon assumption that a linear increase in channel capacity can exponentially reduce signal-to-noise is not so relevant under Simon information. One is better off just trying to find better and higher-level abstractions (i.e., decompositions of problems) to *avoid having to process so much information in the first place*. Limits to an agent's ability to learn to assign credit in complex problems arise instead from limits of the agent's abstractions (i.e., the types of lists and list operations available to an agent) for describing the problem as it relates to (designates) stuff in the world.

You can think of such limits as on the “languages” available to agents for designating and linking lists and their abstractions, rather than one of channel capacity for sending and receiving messages. An organization, Simon observed, does not just impose hierarchy or other structure to channel syntactic messages; it develops a “shared language” among members to communicate meanings that can be mutually understood. Similarly, advances in the intelligence of digital computers depend not just on digital infrastructure (faster chips and more data processing capacity), but also on languages that offer novel abstractions (i.e., coding libraries, and the ability to generate code using natural language).

4.4 “Modularity” as a solution to credit assignment

In 2019, the same year as Sutton's “The Bitter Lesson” blog post, an influential article appeared by Saleema Amershi and her colleagues at Microsoft Research called “Software Engineering for Machine Learning: A Case Study”. The article concerned their observations and experience of how Microsoft organized teams to build machine learning models for using large volumes of data to deploy applications such as recommender systems or computer vision tools at scale.

A main finding was that effective deployment required not just vast chip and data storage capacity to improve individual machine learning models. As Amershi et al. explain, “one cannot (yet) take [a] model of English and add a separate... model for ordering pizza and expect them to work properly together. Similarly, one cannot take that same model for pizza and pair it with an equivalent... model for French and have it work”. Instead, effective deployment depended on organizational processes of interpreting the collective outputs of diverse models that were “**entangled**” in their effects on application performance. Hence, even in machine learning — technologies explicitly adhering to Shannon information — the general problem of learning to assign credit is subsumed within a broader system for semantic information processing, consistent with the Simon information approach.

In the study of AI systems and organizations, having a system learn to assign credit not through brute force, so much as through how problem-solving is abstracted into meaningful modules akin to Simon's watchmaker is called a **modularity** strategy. You should see that a belief in modularity rests on a specific assumption intimately linked to Simon's approach that contrasts with what is implied by Shannon information: that any combinatorial “explosion” in the complexity of problems will *not* be made up for by an exponential increase in the signal-to-noise

ratio produced by a linear increase in channel capacity. More bluntly: adding bigger computers to an AI system will *not* lead to a disproportionately large increase in the ability of the system to learn to assign credit.

5. Dynamics of information processing: IID versus OOD problems

As we have discussed them, the credit assignment mechanisms (brute force in the Shannon information approach; modularity in the Simon information approach) have been mostly static — applying to a particular problem. The Amershi et al. article points to a neglected aspect of our discussion, which is the **dynamics** of processing information across novel problems. Recall from our previous session how in Minsky’s idea of learning in a society, a fundamental challenge of assigning credit arose from the need to deal with novel problem-solving environments — even for problems as seemingly simple as constructing a T-maze out of different materials. To round off our discussion of information as an AI foundation, we need to extend our two approaches to account for such dynamics. A common way to do so in the AI field is to distinguish what are called “IID” and “OOD” problems.

5.1 Shannon information and the IID assumption

To see why the dynamics of information processing are important, let us first draw attention to a subtle yet strong assumption of Shannon information. It assumes an agent has access to a **statistical distribution** of probabilities regarding the possible bit strings of a message. This really means that, even though the agent may face uncertainty about the exact content of the message, it has perfect certainty that the bit strings will be *something* within some set of possible messages. For example, when you prompt an LLM in English to predict the next word of a sentence, we assume that the LLM has perfect certainty that it will be some word in English, and not another language or another kind of symbol (i.e., a soundwave or a picture) altogether. In this sense, Shannon information implies *the world stays basically the same* — or, more formally, *the set of possible outcomes (i.e., messages) have a consistent statistical distribution*.

In statistics, this is called the assumption that the outcomes of a function are **independently and identically distributed**, or “IID”. “Independent” means that outcomes are not interdependent. “Identically distributed” means that the distribution of outcomes — the probabilities of getting each of the possible outcomes — does not change over time. These two ideas capture the basic idea in statistics of “random sampling”. A canonical example is a coin toss, assuming a fair coin. The first coin flip does not depend on the second coin flip, and vice versa. Therefore, the coin flips are *independent*. Further, the probability of heads or tails is both 50%, and these chances in theory should never change, no matter how many times you flip the coin. Therefore, the probabilities in the coin flips are *identically distributed*.

Now consider another example of a coin toss where other factors may come into play. The coin may not be fair (it may be weighted more on one side); the coin may have a wide edge such that there is a chance it will land on the edge; the coin toss may take place over deep water, where there may be no way to actually read the result after the coin sinks to the bottom of the water. These and many other possible violations of the basic IID coin toss set-up can be thought of as **out-of-distribution**, or “**OOD**”. In coin tosses, such possible violations may be unlikely, but in many or most real-world settings, they are the rule rather than the exception, and this was precisely Minsky’s point. To go back to our maze example, Minsky might say it is ultimately more apt to frame the credit assignment problem in AI as: *how to navigate any maze of any size, where the configuration and materials of the maze may change at any time, where even the goal of navigating the maze may be initially unclear, and where you might even need to understand the maze first and how to build it.*

5.2 How to process information about OOD problems?

It may seem that, if credit assignment in AI is ultimately about OOD problems, then the Simon information approach is the only valid one. It makes no assumption that an agent has access to a stable statistical distribution of outcomes for a problem. Yet there are at least two complications; we only touch on these here, but you should keep them in mind as we explore diverse ideas related to both approaches to information throughout the semester.

First, Simon information assumes all problems can be described in terms of lists as the fundamental informational units. Whether that is adequate for processing information across OOD problems, however, is an empirical question. In Simon’s famous watchmaker parable, the watch is given — just like under Shannon information. What if the watchmaker has to assemble many different kinds of watches? Minsky (and many others in the AI field) argue that, to “learn in a society”, or through some other similar way, agents need to process units of information that are often *more complex than lists*, and of *many types*. Consider, for instance, how programming languages today have many types of information (called “data types”), such as dictionaries, tuples, functions, and objects. Perhaps in order to learn to solve OOD problems, some diverse set of information types is a minimal requirement. We will consider such ideas later in the course.

Second, in the total opposite direction of Minsky, could we think about Shannon information differently, such that reducing all information to bit strings is in fact a perfectly natural approach to solving OOD problems — indeed, the only imaginable way? Consider if we think of the IID versus OOD distinction as really only applying to the outcomes of problems that we describe using complex symbols (lists or otherwise), but which we *also* ultimately describe these symbols in a computer in terms of bit strings. In this case, we might assume a statistical distribution *only among the bit strings*, and make predictions about these bit strings that will then generate more complex symbols that we do *not* need to assume follow such distribution. Consider prompting an LLM in English to solve an OOD problem. The prompt that you give it may be a problem that is

radically different from any one it has been given before; yet the English words in the text of your prompt, describable in bit strings, may still be more or less consistent with the overall distribution of English words that the LLM has processed before.

6. Conclusion

In current AI discourse, it is common to relate ideas closer to Shannon information as belonging to “cutting edge” machine learning technologies, while those closer to Simon information are related to supposedly “old-fashioned” AI research. The former is taken to mean an AI system designed to learn with a minimum of “hand-coded” information. The latter is taken to be the opposite, where information is carefully crafted by the designers of the system in advance. In this course, we will look at this way of partitioning the AI field as historically inaccurate and grossly misleading. We will view approaches to information in the study of AI and organizations as an ongoing debate about ways of learning to assign credit, rather than a progression from “old-fashioned” to “cutting edge”. As this week’s course notes have illustrated, a better way to partition the AI field is really more like the distinction between physics and biology — it is about what one considers to be the fundamental units of information (e.g., more atomic bits, or more aggregate lists) for learning to assign credit intelligently. Through this week’s lecture notes, you should see that such assumptions are at once subtle, consequential, and controversial.